

Geometry: Solving Parallelograms — Vol. 2

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DIRECTIONS

In parallelogram ABCD, O is the intersection of the diagonals. Use parallelogram properties to solve for x, y, and missing measures. Show all work.

- 1 Consecutive Angles — Solve for x and find both angles:

$$\angle A = (5x - 8)^\circ, \quad \angle D = (3x + 12)^\circ$$

Answer: _____

- 2 Opposite Sides — Solve for x and find the side length:

$$AB = 4x - 3, \quad CD = 2x + 11$$

Answer: _____

- 3 Diagonals Bisect — Solve for x and find AC:

$$AO = 3x + 4, \quad OC = 5x - 8$$

Answer: _____

- 4 Opposite Angles — Solve for x and find $m\angle P$:

$$\angle P = (6x - 10)^\circ, \quad \angle R = (4x + 20)^\circ$$

Answer: _____

- 5 Find x and all side lengths. Perimeter = 36:

$$AB = 3x + 2, \quad BC = x + 4$$

Answer: _____

- 6 Diagonals Bisect — Solve for x and find BD:

$$BO = 2x + 5, \quad OD = 4x - 7$$

Answer: _____

- 7 Consecutive Angles — Find x and all four angle measures:

$$\angle A = (3x + 6)^\circ, \quad \angle B = (2x + 14)^\circ$$

Answer: _____

- 8 Opposite Sides — Solve for x and y:

$$AB = 3x + y = 16, \quad AD = 2x - y = 4$$

Answer: _____

- 9 Both Diagonals Bisect — Solve for x and y:

$$AO = 2x + y, \quad OC = 4x - 3, \quad BO = 3y - 2, \quad OD = y + 8$$

Answer: _____

- 10 Consecutive Angles — Find x and all four angle measures:

$$\angle A = (4x + 5)^\circ, \quad \angle B = (x + 15)^\circ$$

Answer: _____

- 11 Find x and all side lengths. Perimeter = 54:

$$AB = 4x + 1, \quad BC = 3x - 2$$

Answer: _____

- 12 Opposite Angles — Find x and all four angle measures:

$$\angle A = (7x - 5)^\circ, \quad \angle C = (5x + 15)^\circ$$

Answer: _____



Answer Key & Solutions

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TEACHER NOTES

P1: Opp. sides equal. P2: Opp. angles equal. P3: Consecutive angles sum to 180° . P4: Diagonals bisect each other ($AO=OC$, $BO=OD$). Perimeter= $2(AB+BC)$.

- 1** Consecutive Angles — Solve for x and find both angles:

$$\angle A = (5x - 8)^\circ, \quad \angle D = (3x + 12)^\circ$$

$$= x = 22, \quad m\angle A = 102^\circ, \quad m\angle D = 78^\circ$$

$$\angle A + \angle D = 180^\circ: (5x - 8) + (3x + 12) = 180 \rightarrow 8x = 176 \rightarrow x = 22.$$

- 2** Opposite Sides — Solve for x and find the side length:

$$AB = 4x - 3, \quad CD = 2x + 11$$

$$= x = 7, \quad AB = CD = 25$$

$$\text{Opposite sides equal: } 4x - 3 = 2x + 11 \rightarrow 2x = 14 \rightarrow x = 7.$$

- 3** Diagonals Bisect — Solve for x and find AC :

$$AO = 3x + 4, \quad OC = 5x - 8$$

$$= x = 6, \quad AC = 44$$

$$AO = OC: 3x + 4 = 5x - 8 \rightarrow 12 = 2x \rightarrow x = 6. \quad AO = 22, \quad AC = 2(22) = 44.$$

- 4** Opposite Angles — Solve for x and find $m\angle P$:

$$\angle P = (6x - 10)^\circ, \quad \angle R = (4x + 20)^\circ$$

$$= x = 15, \quad m\angle P = m\angle R = 80^\circ$$

$$\text{Opposite angles equal: } 6x - 10 = 4x + 20 \rightarrow 2x = 30 \rightarrow x = 15.$$

- 5** Find x and all side lengths. Perimeter = 36:

$$AB = 3x + 2, \quad BC = x + 4$$

$$= x = 3, \quad AB = CD = 11, \quad BC = AD = 7$$

$$2(AB + BC) = 36 \rightarrow AB + BC = 18. \quad (3x + 2) + (x + 4) = 18 \rightarrow 4x = 12 \rightarrow x = 3.$$

- 6** Diagonals Bisect — Solve for x and find BD :

$$BO = 2x + 5, \quad OD = 4x - 7$$

$$= x = 6, \quad BD = 34$$

$$BO = OD: 2x + 5 = 4x - 7 \rightarrow 12 = 2x \rightarrow x = 6. \quad BO = 17, \quad BD = 2(17) = 34.$$

- 7** Consecutive Angles — Find x and all four angle measures:

$$\angle A = (3x + 6)^\circ, \quad \angle B = (2x + 14)^\circ$$

$$= x = 32, \quad \angle A = \angle C = 102^\circ, \quad \angle B = \angle D = 78^\circ$$

$$\angle A + \angle B = 180^\circ: 5x + 20 = 180 \rightarrow x = 32. \quad \angle A = 102^\circ, \quad \angle B = 78^\circ. \quad \angle C = \angle A, \quad \angle D = \angle B.$$

- 8** Opposite Sides — Solve for x and y :

$$AB = 3x + y = 16, \quad AD = 2x - y = 4$$

$$= x = 4, \quad y = 4$$

$$\text{Add equations: } 5x = 20 \rightarrow x = 4. \quad \text{Sub back: } 3(4) + y = 16 \rightarrow y = 4.$$

- 9** Both Diagonals Bisect — Solve for x and y :

$$AO = 2x + y, \quad OC = 4x - 3, \quad BO = 3y - 2, \quad OD = y + 8$$

$$= x = 4, \quad y = 5$$

$$BO = OD: 3y - 2 = y + 8 \rightarrow y = 5. \quad AO = OC: 2x + 5 = 4x - 3 \rightarrow x = 4.$$

- 10** Consecutive Angles — Find x and all four angle measures:

$$\angle A = (4x + 5)^\circ, \quad \angle B = (x + 15)^\circ$$

$$= x = 32, \quad \angle A = \angle C = 133^\circ, \quad \angle B = \angle D = 47^\circ$$

$$\angle A + \angle B = 180^\circ: 5x + 20 = 180 \rightarrow x = 32. \quad \angle A = 133^\circ, \quad \angle B = 47^\circ.$$

- 11** Find x and all side lengths. Perimeter = 54:

$$AB = 4x + 1, \quad BC = 3x - 2$$

$$= x = 4, \quad AB = CD = 17, \quad BC = AD = 10$$

$$2(AB + BC) = 54 \rightarrow AB + BC = 27. \quad (4x + 1) + (3x - 2) = 27 \rightarrow 7x = 28 \rightarrow x = 4.$$

- 12** Opposite Angles — Find x and all four angle measures:

$$\angle A = (7x - 5)^\circ, \quad \angle C = (5x + 15)^\circ$$

$$= x = 10, \quad \angle A = \angle C = 65^\circ, \quad \angle B = \angle D = 115^\circ$$

$$\angle A = \angle C: 7x - 5 = 5x + 15 \rightarrow 2x = 20 \rightarrow x = 10. \quad \angle B = 180^\circ - 65^\circ = 115^\circ.$$

